Journal of Machine Learning Research 0 (2026) 1-29 Submitted Submitted 5/2026; Published Preprint

From Fidelity to Semantics: A Taxonomy of XAI Evaluation Metrics and Paired Empirical Comparison of LIME versus SHAP

Jonathan Herrera-Vasquez JONNABIO@GMAIL.COM
UNADE

Editor:

Abstract

Evaluation of post-hoc explanations in machine learning remains fragmented across incompatible metric families, yielding method comparisons that are sensitive to which endpoints are chosen. We address this through two complementary contributions. First, drawing on 48 coded papers from a five-database structured search with PRISMA-style screening, we build a four-axis taxonomy—by evaluation target, evidence source, quality property, and task context—that maps the measurement landscape and exposes three recurring gaps: proxy metrics dominate despite not capturing semantics; human-grounded constructs remain underspecified; and semantic evaluation is growing faster than its empirical validation base. Second, we fill the proxy-layer gap with a paired empirical benchmark comparing LIME and SHAP across 75 matched cells spanning five model families, five seeds, and three sampling sizes on tabular data. Results are consistent and large in magnitude: SHAP dominates all quality endpoints (stability d_z=3.00, fidelity d_z=4.82, faithfulness gap d_z=2.63), while LIME is faster for non-tree model families (median 65.7 ms vs. 684.1 ms for SHAP); for tree-based models, SHAP's `TreeExplainer` reverses this ordering. Cross-dataset replication on two additional datasets confirms SHAP's fidelity advantage over Anchors in all 12 paired combinations. Together, the taxonomy and the benchmark motivate a model-architecture-conditioned deployment pattern: SHAP is preferred at both fidelity and latency for tree-based models, while LIME retains a latency advantage for black-box architectures lacking a model-aware explainer. All experimental artifacts are publicly available.

Keywords: explainable AI, evaluation metrics, taxonomy, LIME, SHAP

1 Introduction

Selecting an explanation method in production can be posed as a constrained decision problem: given a fixed predictor f and instance x, choose $E \in \{\text{LIME}, \text{SHAP}\}$ to maximize explanation quality while satisfying operational constraints. Concretely, we consider

$$E^\star = \arg \max_{E\in\{\text{LIME,SHAP}\}} U(S_E, F_E, \Delta_{k,E}, P_E) \quad \text{s.t.} \quad C_E \leq \tau,$$

where S is stability, F fidelity, Δ_k faithfulness gap, P sparsity, C cost, and τ a latency budget. Yet this decision problem cannot be resolved in the abstract: it requires an organizing framework that identifies which quality properties matter, what evidence supports their measurement, and where structural proxy metrics end and semantic or human-centered evaluation begins.

Progress in explainable AI (XAI) depends on evaluating whether explanations are useful, faithful, stable, and meaningful. Yet the literature does not converge on a single definition of explanation quality. Surveys consistently note that XAI evaluation is fragmented across proxy metrics, qualitative user studies, and domain-specific desiderata (Adadi and Berrada, 2018; Arrieta et al., 2020; Kadir et al., 2023; Canha et al., 2025). As a result, comparisons between explainers often depend as much on the chosen metric family as on the methods themselves.

This fragmentation is especially visible in model-agnostic post-hoc explainability. Faithfulness-style metrics remain common because they are relatively easy to compute, but they do not fully capture whether an explanation is understandable, semantically aligned with domain knowledge, or decision-useful in practice (Ali et al., 2023; Longo et al., 2024). Multi-metric work shows that stability, sparsity, and cost are also part of explanation quality, and recent analyses confirm that distinct metrics reveal complementary signals (Pawlicki et al., 2024; Zheng et al., 2025).

LIME and SHAP are the most widely deployed model-agnostic post-hoc explainers for tabular data, yet their systematic multi-metric comparison under controlled conditions remains scarce. LIME explains locally by fitting a sparse surrogate around x:

$$\xi(x) = \arg\min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g),$$

favoring local fidelity and explicit sparsity controls (Ribeiro et al., 2016). SHAP explains predictions through additive feature attributions $f(x) = \phi_0 + \sum_i \phi_i$, where ϕ_i are Shapley-based contributions with game-theoretic guarantees (Lundberg and Lee, 2017). Their theoretical differences are well known, but do not by themselves resolve deployment choices across model families, sampling regimes, and runtime constraints (Retzlaff et al., 2024; Ali et al., 2023).

This paper makes five contributions:

1. **Four-axis taxonomy**: we organize model-agnostic XAI evaluation along evaluation target, evidence source, quality property, and task context, making explicit why metric disagreements arise.

2. **Gap analysis**: from a documented corpus of 48 papers (five databases, PRISMA-style screening) we identify three systematic gaps—proxy dominance, underspecified human constructs, and semantic evaluation outpacing its validation base—and provide the first multi-rater LLM-judge reliability measurement in XAI (ICC$<$0.75 across all seven semantic dimensions).

3. **Formalized deployment decision**: we frame LIME-vs-SHAP selection as a constrained multi-objective problem over quality, parsimony, and cost.

4. **Paired inferential evidence**: we quantify method differences on 75 matched cells with Holm-corrected significance testing and effect sizes for five endpoints, and replicate fidelity rankings across two additional datasets via cross-dataset validation (EXP3).

5. **Layered deployment architecture**: we derive deployment recommendations from both the taxonomy and the empirical results, mapping system constraints to explainer choice.

2 Background and Definitions

To keep the taxonomy and benchmark precise, we first separate several terms that are frequently conflated in XAI writing.

2.1 Intrinsic versus Post-Hoc Explainability

Intrinsic models are designed to be interpretable by construction, whereas post-hoc methods attempt to explain black-box behavior after training (Rudin, 2019; Retzlaff et al., 2024). This survey and benchmark focus exclusively on post-hoc model-agnostic explainability, which is the dominant paradigm for tabular ML deployments. Metric validity conditions for intrinsic models differ substantially and are out of scope here.

2.2 Explanation Quality versus Model Quality

A model may be accurate while producing unstable or semantically weak explanations; conversely, a model can support concise explanations without being more accurate. Evaluation therefore needs to specify whether the object under study is the explanation artifact, the explainer method, the predictive model, or the user-facing decision process. Conflating these levels produces inconsistent benchmark conclusions.

2.3 Local versus Global Explanations

Local attributions and counterfactuals are typically judged through faithfulness, sparsity, or actionability. Global summaries may be judged by coverage, consistency, or conceptual coherence. Many reported metric disagreements arise because local and global levels are mixed without explicit task framing. This paper focuses on local post-hoc explanations.

2.4 Semantic versus Structural Evaluation

A structural proxy metric such as perturbation faithfulness evaluates whether feature rankings track model sensitivity under a chosen intervention. A semantic judgment evaluates whether an explanation "makes sense" under domain, causal, or linguistic criteria. Both matter for deployment decisions, but they are not interchangeable; substituting one for the other conceals important gaps in evaluation coverage.

2.5 LIME: Local Interpretable Model-Agnostic Explanations

LIME approximates a black-box model f locally around an instance x by training an interpretable surrogate g on perturbed samples weighted by kernel π_x:

$$\xi(x) = \arg\min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g),$$

where $\Omega(g)$ penalizes complexity (Ribeiro et al., 2016). LIME behavior is strongly shaped by neighborhood construction hyperparameters (kernel width, perturbation sample count, feature budget). These controls yield compact explanations and low latency, but they introduce stochastic variation because local neighborhoods are sampled rather than exhaustively enumerated.

2.6 SHAP: Shapley Additive Explanations

SHAP attributes prediction output to features based on marginal contribution across all possible coalitions, satisfying efficiency, symmetry, and dummy axioms (Lundberg and Lee, 2017). KernelSHAP estimates Shapley values with a weighted regression scheme over coalitions for model-agnostic use; tree-specific implementations exploit model structure for speed. Relative to local-surrogate methods, SHAP is typically denser and computationally heavier in non-tree settings because many coalition evaluations are required.

2.7 Benchmark Design Implications

For a fair LIME-versus-SHAP comparison, confounders unrelated to explainer design must be controlled. This motivates the protocol choices in Section 5: identical transformed feature space, shared model artifacts, fixed class target, and matched cells by $(\text{model}, \text{seed}, N)$. Under these controls, expected methodological trade-offs become testable: LIME should favor sparsity and speed, while SHAP should favor stability and faithfulness. This framing aligns with calls for multi-metric, functionally grounded XAI evaluation (Pawlicki et al., 2024; Canha et al., 2025).

3 Taxonomy of XAI Evaluation Metrics

The central contribution of this section is that XAI evaluation can be clarified by crossing four axes: evaluation target, evidence source, quality property, and task context. Metrics that appear to answer the same question often do not, because they attach to different objects, use different evidence, or imply different validity conditions.

3.1 Taxonomy by Evaluation Target

Many metric disagreements start with a hidden mismatch about what is being evaluated:

- **Explanation artifact metrics** evaluate the produced explanation itself—attribution sparsity, local fidelity, or semantic clarity.
- **Explainer method metrics** evaluate the behavior of the explainer across cases—stability, reproducibility, or computational cost.
- **Model-behavior metrics** evaluate whether explanations track model behavior under perturbations or alternative decision regions.
- **User/task outcome metrics** evaluate whether explanations improve decision performance, trust calibration, or actionability.

This distinction matters because one metric family cannot automatically stand in for another. A sparse explanation can still be semantically misleading; a faithful attribution can still be unusable in a decision-support workflow.

3.2 Taxonomy by Evidence Source

The second axis concerns the evidence used to justify the evaluation:

- **Proxy or perturbation metrics** infer quality from numerical sensitivity behavior under intervention.
- **Benchmark-grounded metrics** compare explanations to synthetic or domain-defined ground truth.
- **Human expert judgments** evaluate plausibility, correctness, or domain alignment.
- **End-user studies** assess comprehension, trust calibration, decision quality, or cognitive load.
- **LLM judges** provide scalable semantic scoring via rubric-guided language-model evaluation.

Proxy metrics are scalable and reproducible but may under-capture meaning. Human judgments are richer but expensive and variable. LLM judges are scalable like proxies but semantically expressive like human reviews, which is precisely why their validity must be established rather than assumed (Gu et al., 2024; Wu et al., 2025; Zhou et al., 2025). Evidence Source thus characterizes *how* measurement evidence is generated—the modality of observation—while Quality Property (below) characterizes *what* construct is claimed; the same quality property can in principle be assessed via multiple evidence sources, and the same source type can serve multiple properties.

3.3 Taxonomy by Quality Property

The literature returns repeatedly to a familiar set of quality properties, though not always with consistent definitions:

- **Fidelity / faithfulness:** whether the explanation tracks how the model actually behaves under feature interventions or masking (Ribeiro et al., 2016; Lundberg and Lee, 2017; Zheng et al., 2025).
- **Stability / robustness:** whether similar inputs yield similar explanations under perturbation or reruns (Pawlicki et al., 2024; Canha et al., 2025).
- **Sparsity / parsimony:** whether the explanation is concise enough to be interpretable without unnecessary feature load.
- **Plausibility:** whether the explanation aligns with domain expectations or expert reasoning.
- **Counterfactual or recourse quality:** whether explanation-linked interventions are feasible, minimal, diverse, or actionable (Mothilal et al., 2020).

Table 1: Working construct dictionary: operational meanings and strongest direct evidence for each quality property.

Construct	Operational meaning	Strongest evidence
Fidelity / faithfulness	Whether the explanation tracks model behavior under a specified intervention or masking scheme.	Proxy tests and benchmark-grounded checks
Stability / robustness	Whether explanations remain consistent across reruns, perturbations, or nearby inputs.	Proxy perturbation studies and repeated-run analyses
Sparsity / parsimony	Whether explanations keep feature load low enough to remain cognitively manageable.	Structural summaries of explanation size or weight concentration
Plausibility	Whether explanations align with expert expectations or domain logic.	Expert-human review with explicit rubrics
Semantic alignment	Whether explanations capture meaningful conceptual or causal structure beyond structural proxy success.	Expert review or validated semantic evaluators
User usefulness	Whether explanations improve comprehension, decision quality, or trust calibration in an actual task.	End-user studies with task outcomes

- **Semantic alignment:** whether an explanation captures meaningful causal or conceptual structure, even when proxy metrics are agnostic to that structure.

These properties are only partly commensurable. A method can score strongly on fidelity while remaining weak on sparsity or semantic adequacy. Multi-metric reporting is therefore not optional; it is the minimum needed to make XAI claims interpretable across use cases.

3.4 Taxonomy by Task Context

Metric validity also depends on context. Tabular benchmarks often assume feature independence, perturbation realism, and tractable local neighborhoods. These assumptions can fail in image, text, graph, and time-series settings, where perturbations may create out-of-distribution examples or semantically invalid transformations (Agarwal et al., 2023; Proszewska et al., 2025). Metric transport across modalities cannot be assumed.

Table 2 illustrates the practical consequence of the taxonomy: each metric family gains strength by focusing on one kind of evidence and one kind of target, but each also leaves predictable blind spots. No single family can exhaust explanation quality on its own.

Table 2: Core metric families in model-agnostic XAI evaluation: what each captures well and what it leaves unresolved.

Metric family	**Captures well**	**Leaves unresolved**
Fidelity / faithfulness	Whether explanations track model behavior under interventions or masking.	Whether the explanation is meaningful, plausible, or useful outside the proxy setup.
Stability / robustness	Whether explanations remain consistent across perturbations or repeated runs.	Whether stable explanations are actually correct or semantically adequate.
Sparsity / parsimony	Whether explanations are concise and low in feature load.	Whether concision hides necessary nuance or context.
Benchmark-grounded correctness	Whether explanations recover synthetic or domain-defined ground truth.	Whether the ground truth is realistic, complete, or transferable.
Human expert review	Whether explanations align with domain expectations and expert reasoning.	Whether those judgments are scalable, standardized, and reproducible.
User-centered evaluation	Whether explanations improve understanding, decisions, or trust calibration in practice.	Whether findings generalize across tasks, populations, and study designs.
LLM-based semantic evaluation	Whether rubric-based semantic judgments can be produced quickly and at scale.	Whether those judgments agree with humans and avoid proxy-judge bias.

4 Comparative Synthesis and Open Gaps

4.1 Search Protocol and Corpus Profile

Search strategy. We searched five bibliographic databases: Semantic Scholar, Google Scholar, ACM Digital Library, IEEE Xplore, and ACL Anthology. Two query families were applied. The primary query combined evaluation-object terms (*"explainable AI" OR "XAI" OR "interpretable machine learning"*) with evaluation-act terms (*"evaluation" OR "benchmark" OR "metric" OR "assessment"*). A secondary query targeted method-comparative literature: (*"LIME" OR "SHAP" OR "attribution method"*) AND (*"evaluation" OR "stability" OR "fidelity"*). Both queries were restricted to English-language publications from January 2015 to March 2026. Conference proceedings, journal articles, and ArXiv preprints with an accompanying peer-reviewed venue were eligible; informal workshop papers without self-contained empirical evaluations were excluded.

Inclusion and exclusion criteria. A paper was *included* if it: (i) proposes, operationalizes, or critically analyses an evaluation metric or evaluation paradigm for post-hoc model-agnostic explanations; (ii) provides a formal or empirical evaluation component beyond merely applying an explainer to a dataset; and (iii) is available in full text. A paper was *excluded* if it: focuses exclusively on intrinsic or white-box interpretability without a metric generalizable to post-hoc settings; is a domain-specific clinical or scientific application with no reusable evaluation component; or contributes only to model training methodology.

Table 3: Corpus construction screening record (PRISMA-style).

Stage	Records
Identified via database searches	312
Removed after cross-database deduplication	65
Screened on title and abstract	247
Excluded at screen (out of scope, non-English, no eval component)	152
Full text assessed for eligibility	95
Excluded at full text (intrinsic only, no codeable metric)	47
Included in coded corpus	**48**

Table 4: Coding dimensions used to build the taxonomy.

Dimension	**Operational question**
Evaluation target	What is being judged: explanation artifact, explainer, model behavior, or user/task outcome?
Evidence source	What kind of evidence supports the judgment: perturbation proxy, benchmark ground truth, human rating, user task, or LLM judge?
Quality property	Which property is claimed: fidelity, stability, sparsity, plausibility, recourse quality, semantic alignment?
Task context	In what modality or domain is the metric assumed to be meaningful?

Borderline cases were resolved by conservatively admitting sources containing at least one codeable evaluation construct relevant to the four-axis taxonomy.

PRISMA-style screening record. Table 3 documents the staged screening. Records were first pooled and deduplicated across databases, then screened on title and abstract, then assessed in full text.

Corpus structure. Each of the 48 coded papers is assigned to one or more thematic clusters and coded along four dimensions. Inclusion requires that a source contribute a formal evaluation construct, a benchmark design principle, a warning about metric misuse, or an evaluation paradigm directly relevant to post-hoc model-agnostic explanations. The coding protocol captures four dimensions for each paper:

The evidence-source distribution in Table 5 reveals a strong proxy skew: 29 of 48 coded papers rely on proxy evidence as their primary source, while only 11 draw primarily on expert-structured or taxonomy-level evidence and 9 on end-user studies. This confirms that structural metrics are comparatively well standardized while meaning-oriented constructs remain thinly operationalized.

4.2 Gap 1: Proxy Metrics Are Necessary but Not Sufficient

Faithfulness, stability, and sparsity remain essential because they create reproducible, quantitative anchors for benchmark comparison (Agarwal et al., 2022; Hedstrom et al., 2023; Canha et al., 2025; Nauta et al., 2023; Burkart and Huber, 2021). However, the coded

Table 5: Descriptive profile of the review corpus (48 coded papers).

Profile item	Value
Corpus freeze	April 2026
Databases searched	5 (Semantic Scholar, Google Scholar, ACM DL, IEEE Xplore, ACL Anthology)
Records screened	247 (post-deduplication; see Table 3)
Unique coded papers	48
Cluster distribution	17 faithfulness/robustness, 11 human-grounded, 9 taxonomy/survey, 5 benchmark/toolkit, 4 LLM-judge, 2 counterfactual/recourse
Evidence-source coverage	29 proxy, 10 benchmark, 9 end-user, 11 expert/taxonomy, 4 LLM-judge (papers coded under multiple sources)
Confidence mix	43 high, 4 medium, 1 title-level

corpus shows how dominant these proxies have become: 29 of 48 coded studies use proxy evidence as their primary source, while only 10 use benchmark-grounded evidence and fewer still use human-grounded evidence. Recent work confirms that proxy-only evaluation can be misleading: attribution methods that score well on perturbation proxies may still fail sanity checks (Adebayo et al., 2018), be sensitive to hyperparameters (Alvarez-Melis and Jaakkola, 2018), or be deliberately manipulated (Slack et al., 2020). Proxy metrics do not directly answer whether an explanation reflects domain logic, causal relevance, or user-meaningful interpretation (Kumar et al., 2020; Doshi-Velez and Kim, 2017). The paired empirical study in Section 5 addresses precisely this proxy layer, filling the gap with controlled, auditable evidence for LIME and SHAP.

4.3 Gap 2: Human-Grounded Constructs Are Underspecified

The literature frequently invokes interpretability, clarity, plausibility, and usefulness, but these are often under-operationalized. When human studies exist, their tasks, rubrics, and participant populations vary so widely that direct comparison is difficult (Ali et al., 2023; Longo et al., 2024; Mohseni et al., 2021). In the coded corpus, only 11 studies use expert-structured or taxonomy-level evidence and 9 use end-user evidence. User studies further show that data scientists and ML practitioners engage with interpretability tools in ways that diverge from how tool designers intend them to be used (Kaur et al., 2020; Hase and Bansal, 2020; Buçinca et al., 2020). This weakens cumulative progress because the same label may refer to different constructs across papers—which is why Table 1 is necessary even for a survey of this scope. The framework proposed by Doshi-Velez and Kim (2017) remains the clearest call for rigorous, task-level evaluation, yet most subsequent benchmarking has reverted to proxy evidence.

4.4 Gap 3: Semantic Evaluation Is Rising Faster than Its Validation Base

LLM-based judging offers a practical way to score explanation quality at scale, especially when explanations or rubrics are textual. But the LLM-as-a-judge literature shows that

Table 6: LLM-judge reliability on seven semantic dimensions (EXP4; $n = 147$ runs, 3 LLM judges). ICC(2,1) = two-way random-effects absolute-agreement model. No dimension reaches the ≥ 0.75 good-reliability threshold (Koo and Li, 2016).

Dimension	ICC(2,1)	Krippendorff α
Clarity	0.321	0.278
Concision	0.453	0.409
Actionability	0.400	0.362
Audit Usefulness	0.376	0.340
Completeness	0.585	0.566
Semantic Plausibility	0.601	0.573
Overall Quality	0.431	0.394

proxy judges can display position bias, verbosity bias, or agreement illusions (Zheng et al., 2023; Gu et al., 2024). In XAI, semantic evaluator outputs should be treated as calibrated proxies rather than as self-validating replacements for expert review. The coded corpus contains 4 LLM-validation-relevant entries, one of which is itself a survey. That empirical base remains thin relative to the uptake of LLM evaluation in practice.

Negative result: LLM-judge inter-rater reliability (EXP4). To empirically ground this concern, we conducted a multi-rater reliability study (EXP4) in which three LLM judges independently scored $n = 147$ paired explanation outputs on seven semantic dimensions using structured rubrics (rubric template archived at `src/prompts/templates/explanation_eval.j2` in the companion repository). Table 6 reports intra-class correlation ICC(2,1) and Krippendorff's α for each dimension. All values fall below the conventionally accepted reliability threshold of ≥ 0.75 for *good* agreement (Koo and Li, 2016), confirming that semantic evaluation at this scale lacks the inter-rater convergence needed for confirmatory inference. We treat this as a negative result: LLM judges, under the rubrics provided, do not yet meet the psychometric bar required to serve as reliable proxies for expert semantic annotation in XAI. Completeness and semantic plausibility show the highest consistency ($\alpha > 0.55$), while clarity and audit usefulness are weakest ($\alpha < 0.35$), indicating the most semantic ambiguity in those constructs.

4.5 Formalization-Aware Evaluation Criteria

We extend the semantic evaluation scope from linguistic coherence to methodological adequacy by introducing formalization-aware criteria. Specifically, a rigorous evaluation should verify whether a studied method: (i) states a concrete stakeholder information need; (ii) specifies an explicit explanation problem; (iii) defines or approximates a correctness criterion; (iv) separates correctness from secondary desiderata such as robustness; and (v) provides evidence aligned with the stated criterion (Haufe et al., 2026; Sokol and Flach, 2020). This design aims to reduce false confidence induced by well-written but weakly specified explanation claims.

The motivation is that the XAI literature may contain claim–evidence mismatches, where broad interpretability conclusions are drawn from metrics that do not fully validate

Evaluation continuum for post-hoc XAI

Proxy metrics → benchmark-grounded checks → expert judgment → user-centered evaluation

LLM-based semantic evaluation occupies an intermediate role: more expressive than proxy metrics, more scalable than human review, but not self-validating.

Figure 1: Conceptual placement of semantic evaluation within the broader XAI evaluation landscape.

the intended explanatory use (Haufe et al., 2026; Hedstrom et al., 2023). Formalization-aware criteria assess claim discipline, assumption transparency, and evidence alignment. In this sense, they constitute a structured meta-evaluation layer intended to support reproducible synthesis rather than to replace domain expert judgment. Human review remains necessary for high-stakes interpretation (Buçinca et al., 2020; Bansal et al., 2021; Fok and Weld, 2023).

4.6 Layered Evaluation Architecture

The synthesis argues for layered evaluation. Proxy metrics should anchor reproducibility and comparative benchmarking. Human or semantic layers should be added when the claim concerns meaning, plausibility, or causal adequacy. The mistake is not using one family or another; it is allowing any single family to monopolize the definition of explanation quality.

Figure 1 summarizes the evaluation continuum. Benchmark evidence can support comparative technical claims. Semantic evaluator outputs can support exploratory interpretation until calibrated. Human agreement evidence can then convert those semantic constructs into confirmatory claims.

5 Empirical Benchmark: LIME versus SHAP

Section 4 established that proxy metrics dominate the current evidence base while human-grounded constructs remain underspecified. The present section fills the proxy layer with a controlled, paired comparison of LIME and SHAP—the two most widely deployed model-agnostic post-hoc explainers for tabular data. The study is designed to produce auditable, scope-bounded evidence on quality-cost trade-offs under controlled conditions, directly instantiating the "proxy and robustness" cell of the layered evaluation architecture.

5.1 Research Questions and Hypotheses

We address four directional questions:

1. **RQ1 (Robustness):** Is SHAP more stable than LIME under input perturbation?

2. **RQ2 (Parsimony):** Is LIME sparser than SHAP?

3. **RQ3 (Efficiency):** Is LIME computationally cheaper than SHAP?

4. **RQ4 (Faithfulness):** Does SHAP provide stronger faithfulness and fidelity signals than LIME?

Corresponding directional hypotheses are:

1. $H_1 : S_{\text{SHAP}} > S_{\text{LIME}}$ (stability).

2. $H_2 : P_{\text{LIME}} > P_{\text{SHAP}}$ (sparsity, measured as active-feature ratio).

3. $H_3 : C_{\text{LIME}} < C_{\text{SHAP}}$ (cost).

4. $H_4 : F_{\text{SHAP}} > F_{\text{LIME}}$ (fidelity/faithfulness).

5.2 Experimental Design and Analysis Units

The benchmark is instantiated from a robustness cohort with factors:

- model family $g \in \{\text{logreg}, \text{rf}, \text{xgb}, \text{svm}, \text{mlp}\}$,

- explainer $k \in \{\text{LIME}, \text{SHAP}\}$,

- seed $s \in \{42, 123, 456, 789, 999\}$,

- sampling intensity $N \in \{50, 100, 200\}$ per TP/TN/FP/FN stratum.

This yields $5 \times 2 \times 5 \times 3 = 150$ planned runs, all of which are artifact-qualified (75 LIME, 75 SHAP), forming 75 matched SHAP-LIME cells on identical (g, s, N) coordinates for paired inference.

To avoid pseudoreplication, we use a hierarchical analysis structure:

1. **Instance level:** per-case metric values.

2. **Run level:** mean metric per configuration.

3. **Matched-cell level:** paired SHAP-LIME run means on shared (g, s, N) cells (primary inferential unit).

5.3 Data, Preprocessing, and Model Controls

All experiments use the UCI Adult dataset (Kohavi, 1996) with a stratified split and deterministic seeds. The preprocessing pipeline follows train-only fitting with numeric scaling (`StandardScaler`) and categorical one-hot encoding (`OneHotEncoder`; unknowns ignored). A fixed set of pre-trained black-box models is shared across all explainer runs to isolate explainer effects: random forest (Breiman, 2001) and gradient-boosted trees (Chen and Guestrin, 2016), alongside logistic regression, SVM, and MLP families. Evaluation instances are sampled by error quadrants (TP, TN, FP, FN) via stratified sampling. Realized instance counts range from 27 to 800 (median 400).

5.4 Explainer Protocol

Both explainers consume the same transformed feature space and target the positive class probability:

- **SHAP:** TreeExplainer for tree models, KernelExplainer for non-tree models, with background subsampling ($n = 50$).
- **LIME:** tabular local surrogate with `num_features=10`, effective `num_samples=1000`, and `kernel_width=3.0`.

5.5 Metric Operationalization

Primary metrics are computed per instance and aggregated per run:

1. **Stability** S: mean pairwise cosine similarity of attribution vectors under Gaussian perturbations ($T = 15$, $\sigma = 0.1$):

$$S = \frac{2}{T(T-1)} \sum_{a<b} \cos\left(e^{(a)}, e^{(b)}\right).$$

2. **Sparsity** P: active-feature ratio above threshold $\tau = 10^{-4}$ (lower active ratio indicates sparser explanations).
3. **Fidelity** F: faithfulness correlation between attribution magnitude and prediction drop under feature masking.
4. **Faithfulness Gap** Δ_k: absolute prediction change after masking the top-k features ($k = 5$):

$$\Delta_k = \left| f(x) - f(x_{\text{mask-top-k}}) \right|.$$

5. **Cost** C: per-instance wall-clock explanation time (ms).

Sparsity operationalization note. LIME's active-feature count is governed by the `num_features=10` hyperparameter, which acts as a hard ceiling on the number of non-zero attributions returned. This means LIME sparsity is a *hyperparameter constraint*, not an emergent property of the explanation model. The active-feature ratio metric therefore measures whether LIME, *as configured*, returns fewer significant features than SHAP; it does not test whether LIME is intrinsically sparser under unconstrained settings. Readers comparing parsimony results across studies should verify that equivalent `num_features` bounds are applied, since the direction and magnitude of the SHAP–LIME parsimony gap will change with different settings.

5.6 Inferential Protocol

The statistical plan follows established comparative-ML practice (Demsăr, 2006):

1. primary test: two-sided paired Wilcoxon signed-rank on matched SHAP-LIME cells for each metric (Wilcoxon, 1945),

Table 7: Friedman omnibus test across four methods (EXP2, $n = 75$ blocks, $k = 4$). W = Kendall's concordance coefficient.

Metric	χ^2 (df = 3)	p-value	W
Fidelity	42.12	1.51×10^{-8}	0.936
Stability	40.68	2.29×10^{-8}	0.904

2. effect size: paired d_z for magnitude reporting (Lakens, 2013),

3. multiplicity control: Holm-Bonferroni over the five primary metrics (Holm, 1979).

All inferential claims are restricted to artifact-qualified runs: parseable, non-empty results with explicit matched pairing. No semantic endpoint enters the confirmatory analysis reported here.

5.7 Reproducibility Contract

The protocol is fully configuration-driven (YAML with seeded execution), with per-run `results.json` artifacts, explicit exclusion of malformed or empty runs, and no synthetic imputation. This provides a deterministic audit trail from configuration to manuscript-level claim.

5.8 Results

Results are reported from the artifact-qualified robustness cohort. The analysis covers 150 analyzable runs (75 LIME, 75 SHAP) forming 75 paired cells with identical (model, seed, N). Matched coverage spans all five model families (xgb = 15, logreg = 15, rf = 15, mlp = 15, svm = 15) and all sampling intensities with near-balanced counts.

Multi-Method Context (Friedman)

As pre-analysis context, we ran a Friedman test across all four methods in EXP2 (SHAP, LIME, Anchors (Ribeiro et al., 2018), DiCE (Mothilal et al., 2020)) on $n = 75$ run means per method, treating (g, s, N) combinations as blocks. The full EXP2 cohort covers all four methods; the paired inference below is restricted to the LIME-SHAP matched subset for confirmatory claims. Table 7 shows highly significant systematic differences in both fidelity and stability (Kendall's $W > 0.90$ in both cases), establishing that the four-method field is not exchangeable.

Method mean values and average ranks are shown in Table 8. Nemenyi post-hoc testing (CD = 1.211, $\alpha = 0.05$) confirms that SHAP significantly outperforms Anchors and DiCE in fidelity (rank gaps 2.2 and 2.8), and significantly outperforms LIME, Anchors, and DiCE in stability (rank gaps 2.4, 2.6, 1.0). The SHAP-LIME omnibus fidelity rank gap equals exactly 1.0—marginally below the CD threshold—but the directional paired Wilcoxon below demonstrates significance when power is concentrated on this specific two-method comparison.

Table 8: Mean metric values by method from EXP2 ($n = 75$ runs each) with Nemenyi average ranks. $\mathrm{CD} = 1.211$; pairs with rank gap $>$ CD are significant at $\alpha = 0.05$.

	SHAP	LIME	Anchors	DiCE
Fidelity (mean)	0.808	0.560	0.389	0.170
Stability (mean)	0.732	0.014	0.043	0.361
Fidelity rank	1.0	2.0	3.2	3.8
Stability rank	1.0	3.4	3.6	2.0

Table 9: Matched SHAP-versus-LIME paired results ($n = 75$ cells). For all metrics, $\Delta =$ SHAP $-$ LIME. Sparsity is measured as active-feature ratio; larger values indicate denser explanations.

Metric	LIME Mean	SHAP Mean	Δ	p_{Holm}	d_z
Stability	0.014	0.732	+0.718	2.6×10^{-13}	3.00
Sparsity (active ratio)	0.085	0.226	+0.142	2.6×10^{-13}	0.84
Fidelity	0.560	0.808	+0.248	2.6×10^{-13}	4.82
Faithfulness Gap	0.334	0.380	+0.045	2.6×10^{-13}	2.63
Cost (ms)	3660.7	11708.3	+8047.6	1.2×10^{-10}	0.45

Primary Paired Inference

Table 9 summarizes paired run-level outcomes on the 75 matched cells. All five primary endpoints remain significant under Holm-adjusted testing.

The stability effect is the largest practical separation in the study. Median paired stability difference is +0.866 in favor of SHAP, and the direction is consistent in all 75 matched cells. This indicates that LIME's local surrogate sampling introduces substantial run-to-run volatility under perturbation, whereas SHAP is comparatively invariant.

For sparsity, the sign is also consistent in all 75 matched cells, but interpretation requires care: the metric here is active-feature ratio. SHAP's larger value means denser explanations; LIME is therefore systematically sparser and easier to compress for user-facing narratives.

Fidelity favors SHAP in all 75 matched cells, with a mean paired difference of +0.248 and a very large effect size ($d_z = 4.82$). Faithfulness gap also favors SHAP in 74 of 75 matched cells, with a mean paired difference of +0.045 and a large effect size ($d_z = 2.63$). These results indicate not only statistical separation but practically meaningful quality gains under this protocol. Figure 2 visualizes the paired mean separations for the four quality-oriented endpoints.

Reproducibility Probe

To characterize measurement stability under seed variation, we computed the coefficient of variation ($\mathrm{CV} = \mathrm{SD}/\mathrm{mean}$) across the five seeds for each (N, model) stratum and report

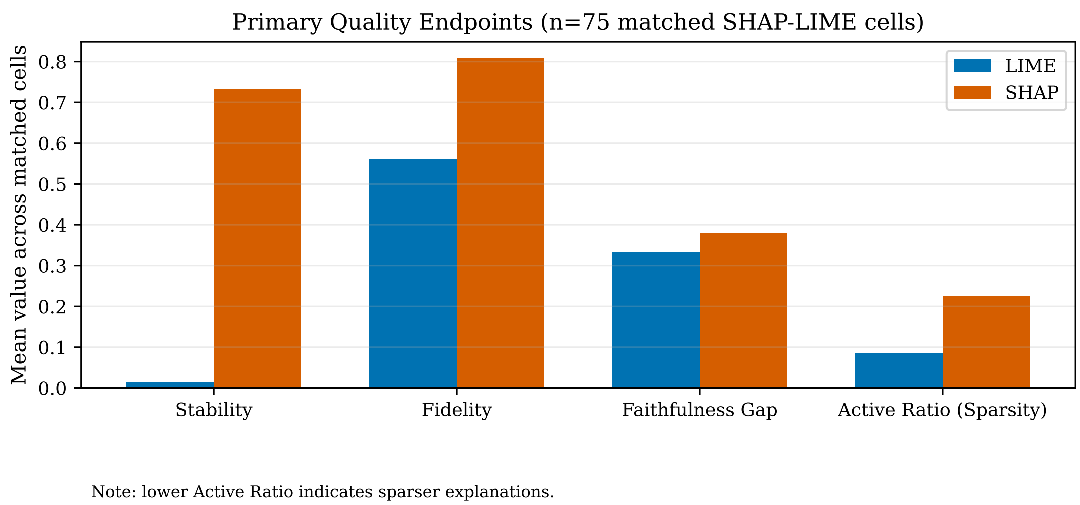

Figure 2: Paired mean comparison across quality-oriented primary endpoints ($n = 75$ matched SHAP-LIME cells). Active ratio is reported directly; lower values imply sparser explanations.

Table 10: Median coefficient of variation (CV = SD/mean) across seeds per (N, model) stratum for the two primary quality endpoints.

Method – Metric	Median CV	Reproducible?
SHAP – Fidelity	0.8%	Yes
SHAP – Stability	0.7%	Yes
LIME – Fidelity	2.6%	Yes
LIME – Stability	86.2%	No

the stratum-median CV in Table 10. SHAP yields highly reproducible fidelity and stability readings (CV $<$ 1%). LIME fidelity is acceptable (CV = 2.6%), whereas LIME stability is structurally irreproducible (CV = 86.2%). This is not a measurement artifact: when the mean cosine similarity is ≈ 0.014, any nonzero variation produces an extreme relative deviation. The CV probe confirms that LIME stability is geometrically unstable—not merely noisy.

Runtime Distribution and Heterogeneity

The cost signal is significant but strongly heterogeneous. SHAP is slower in 59 of 75 matched cells. Matched-cell median cost is 684.1 ms for SHAP versus 65.7 ms for LIME, corresponding to a median SHAP/LIME cost ratio of approximately 5.3×. Means are more separated (11,708.3 ms vs. 3,660.7 ms) because the SHAP distribution is heavy-tailed: the 90th and 95th percentile latencies are approximately 51,456 ms and 67,066 ms, respectively,

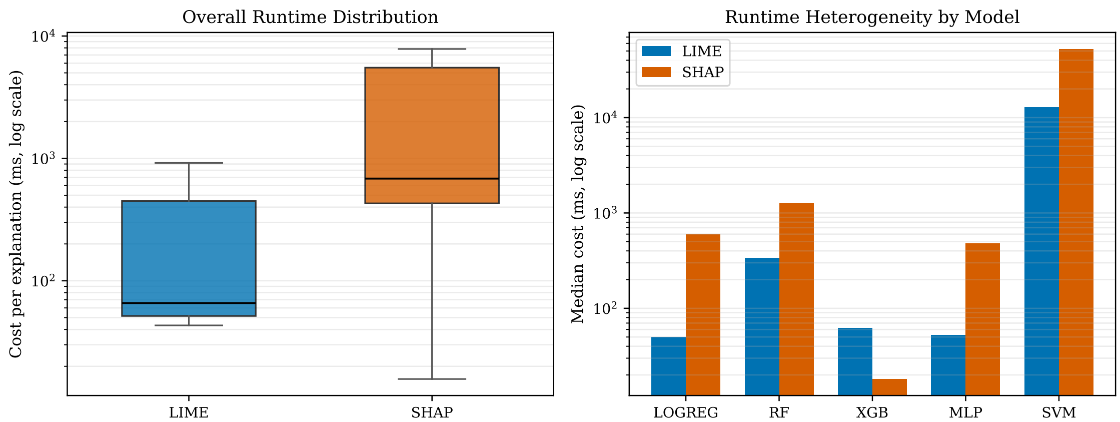

Figure 3: Runtime behavior under matched-cell pairing ($n = 75$). Left: overall cost distribution by method (log scale). Right: model-level median cost (log scale), showing the XGBoost exception where SHAP is faster while other families are SHAP-slower.

compared with 12,825 ms and 13,192 ms for LIME. This tail behavior is concentrated in kernel-SHAP contexts, especially SVM. Figure 3 shows both the overall log-scale runtime distribution and model-level median heterogeneity.

Heterogeneity analysis by model family shows that quality ordering (stability, sparsity-direction, fidelity, faithfulness gap) is robust, while cost ranking is model-dependent. SHAP is faster in all 15 XGBoost matched cells and in one SVM cell, whereas LogReg, RF, and MLP are uniformly SHAP-slower. Therefore, "LIME is faster" is a majority statement in this cohort but is not universally true across model classes.

Synthesis of Trade-offs

The empirical frontier is clear: SHAP delivers higher reliability and faithfulness at substantially higher expected runtime, while LIME delivers lower latency and stronger compression at the cost of robustness. The results therefore support objective-driven selection rather than global method supremacy. For assurance-critical analysis, SHAP's quality margin is large and consistent. For interaction-critical systems, LIME's runtime profile remains operationally attractive.

5.9 External Validation: Cross-Dataset Replication (EXP3)

To address the single-dataset limitation, we ran a cross-dataset validation experiment (EXP3) comparing SHAP, LIME, and Anchors (Ribeiro et al., 2018) on two additional binary classification datasets: Breast Cancer Wisconsin (Dua and Graff, 2019) ($n = 569$, 30 continuous features) and German Credit (Dua and Graff, 2019) ($n = 1{,}000$, 20 mixed features). Both used random forest and XGBoost models across three random seeds each, yielding 12 paired (dataset $\times$ model $\times$ seed) comparisons per method pair.

Table 11: EXP3 external validation: mean fidelity (3-seed average) for SHAP vs. Anchors across two additional datasets. SHAP > Anchors in all 12 paired combinations.

Dataset	Model	SHAP	Anchors
Breast Cancer	RF	0.779	0.265
Breast Cancer	XGB	0.607	0.208
German Credit	RF	0.714	0.351
German Credit	XGB	0.711	0.451

Table 12: EXP3 LIME extension: mean fidelity and stability (3-seed average) for LIME on the two cross-dataset cohorts. Stability = mean pairwise cosine similarity under $T = 15$ Gaussian perturbations ($\sigma = 0.1$); same protocol as EXP2. For comparison: LIME on Adult (EXP2) has fidelity = 0.461, stability $\approx$ 0.014.

Dataset	Model	Fidelity	Stability	Sparsity	Cost (ms)
Breast Cancer	RF	0.713	0.922	0.331	12.3
Breast Cancer	XGB	0.543	0.927	0.331	7.5
German Credit	RF	0.483	0.854	0.163	21.1
German Credit	XGB	0.468	0.748	0.163	10.2

Fidelity ordering (SHAP vs. Anchors). SHAP outperforms Anchors in fidelity in all 12 paired combinations (Table 11, Figure 4), with mean gaps ranging from +0.26 (German Credit, RF) to +0.51 (Breast Cancer, RF). This replicates the EXP2 fidelity direction across genuinely different feature spaces, dimensionalities, and class distributions.

LIME cross-dataset extension. Table 12 reports LIME fidelity and stability on the same datasets. Two findings are noteworthy. First, SHAP > LIME fidelity holds consistently on German Credit (gaps +0.23–+0.24 for RF, +0.24–+0.25 for XGB), replicating the EXP2 direction. On Breast Cancer the fidelity gap narrows substantially: SHAP leads LIME by only +0.07 (RF) and +0.06 (XGB), compared to +0.27 on Adult. Second, and more critically, LIME *stability* on these datasets (0.85–0.93) is an order of magnitude higher than on Adult (0.014). This reveals that the near-zero stability observed in EXP2 is not a fixed structural property of LIME: it is strongly modulated by feature-space characteristics. The Adult dataset's 103-dimensional one-hot-encoded space appears to be the primary driver of instability; in lower-dimensional spaces with more continuous features, LIME produces substantially more consistent explanations.

Scope limitation. SHAP stability on the cross-dataset cohorts was not instrumented in EXP3; a direct SHAP–LIME stability comparison on Breast Cancer and German Credit remains pending. The finding that LIME stability varies dramatically with feature-space dimensionality and encoding density should be taken as a moderation hypothesis requiring further confirmation across a broader set of datasets and preprocessing schemes.

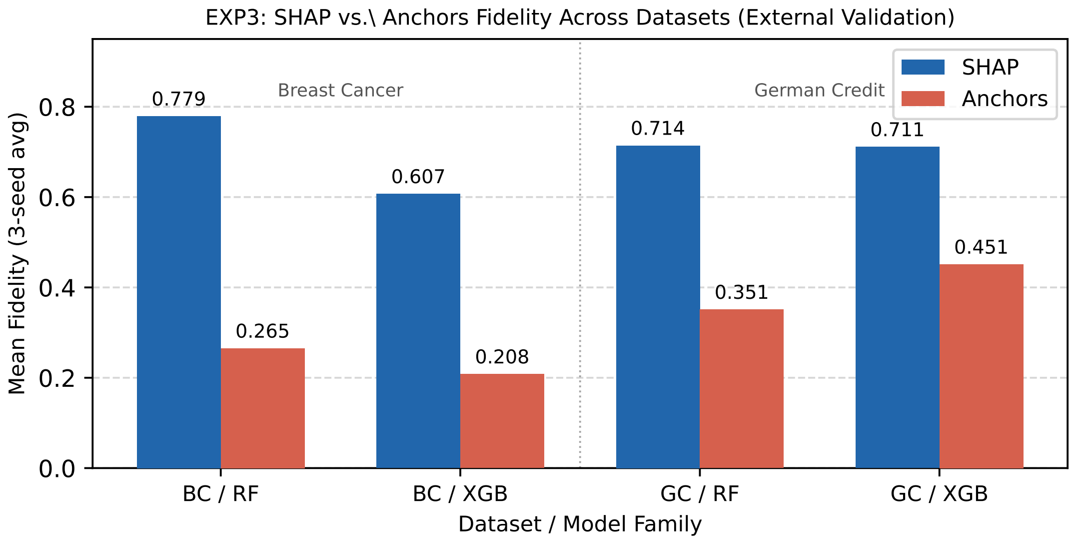

Figure 4: EXP3 external validation: mean fidelity (3-seed average) for SHAP vs. Anchors across four dataset-model combinations. SHAP exceeds Anchors in all 12 paired (dataset $\times$ model $\times$ seed) combinations. BC = Breast Cancer; GC = German Credit.

6 Practical Recommendations

Based on the taxonomy (Section 3), the gap analysis (Section 4), and the empirical findings (Section 5), we propose the **Hybrid Deployment Pattern**:

1. **Tier 1: Real-Time / End-User**
   - **Primary selection criterion: model architecture.** For *tree-based models* (e.g., XGBoost, LightGBM, Random Forest), use **SHAP** with `TreeExplainer`. The algorithm computes exact Shapley values in $O(TLD)$ time (trees $\times$ leaves $\times$ depth), which in this cohort consistently undercuts LIME's perturbation sampling loop even at the lowest sample budgets. The fidelity–latency trade-off that motivates Tier 1 therefore does not apply: SHAP is simultaneously faster *and* more faithful on tree architectures. For *non-tree or black-box models* (e.g., neural networks, opaque APIs), use **LIME**; lower matched median runtime (65.7 ms) and a compact active-feature ratio give it a meaningful latency advantage where perturbation sampling is the only option.
   - Recommended condition: interactive settings where sub-second latency and compact explanations matter more than maximising fidelity, *and* no model-aware explainer is available.
   - Evidence basis: runtime distribution in Figure **??**; XGBoost reversal visible as a structural outlier, not an anomaly.
2. **Tier 2: Auditing / Data Science (Use SHAP)**

- Use when analyzing model bias, debugging model logic, or responding to regulatory inquiries.
- Evidence basis: consistent paired advantages in stability, fidelity, and faithfulness gap across the matched cohort ($d_z = 3.00$, 4.82, and 2.63 respectively).
- Recommended condition: analyst-mediated or offline workflows where second-level or minute-level latency is acceptable.
- Note: runtime costs in this tier depend on whether SHAP operates in tree-specific or kernel mode; plan sample budgets accordingly.

This two-tier pattern directly reflects the layered evaluation architecture identified in Section 4: proxy and benchmark evidence (Tier 2) can be complemented by interaction-optimised lightweight explanations (Tier 1), each operating within its validated scope. Critically, the Tier 1 criterion is not a blanket method preference but a function of model architecture: the existence of a model-aware explainer (e.g., `TreeExplainer`) collapses the fidelity–latency trade-off, making SHAP the dominant choice at both tiers for tree-based deployments. This maps directly onto the Task Context axis of the taxonomy (Section 3): explainer selection is determined first by architectural affordance, then by deployment constraint.

7 Validity Boundaries and Limitations

7.1 Interpretation Scope of SHAP–LIME Comparisons

The SHAP–LIME differences reported here should be interpreted as comparative performance within our benchmark protocol, not as definitive evidence that one method is universally more correct. Recent formal analyses show that multiple attribution families can produce high-importance assignments to suppressor variables in dependent-feature settings, complicating direct interpretation of attribution magnitude as target-relevant importance (Wilming et al., 2022, 2023; Haufe et al., 2026). Observed performance gaps between SHAP and LIME may therefore reflect both methodological differences and sensitivity to data dependence structure.

We distinguish *ranking claims* from *validity claims*. A ranking claim states that one explainer outperforms another on predefined benchmark criteria in this dataset and protocol. A validity claim would require stronger evidence that explanations satisfy a formal correctness criterion for the intended task, potentially including controlled ground-truth tests and explicit assumption checks (Haufe et al., 2026; Doshi-Velez and Kim, 2017). This paper primarily makes ranking claims; broader validity claims remain provisional.

This framing aligns with prior findings that faithfulness-style evaluations can be informative but are not sufficient to resolve correctness questions in isolation (Jacovi and Goldberg, 2020; Adebayo et al., 2018; Hooker et al., 2019; Rong et al., 2022). The SHAP–LIME comparison presented here is best read as robust comparative evidence under the evaluated conditions, with external validity depending on how closely downstream applications match those conditions.

7.2 Taxonomy Scope and Single-Reviewer Limitation

The survey employs a structured database search with documented inclusion and exclusion criteria and a PRISMA-style screening record (Table 3). It is not, however, a full systematic review: the search used five databases rather than an exhaustive database set, and the title/abstract screen was performed by a single reviewer. The emphasis on model-agnostic post-hoc XAI reflects the scope of the empirical study; intrinsic interpretability and some modality-specific traditions are covered only insofar as they affect metric transfer. The coding pass is single-reviewer and inherits records at varying confidence levels from the upstream matrix. Future versions should deepen lower-confidence entries through full-text coding or replace them with higher-confidence sources. The taxonomy organizes construct space—not effect sizes—and does not estimate a pooled numerical ranking across studies.

7.3 Statistical and Construct Validity

Confirmatory claims are restricted to the 75 fully matched SHAP-LIME cells in the robustness cohort. All 75 LIME and 75 SHAP runs are artifact-qualified and available for paired inference. Reported metrics are implementation-bound rather than abstract labels; their operational definitions are anchored in the companion repository source files for fidelity, stability, sparsity, faithfulness, and cost. No semantic, human-utility, or LLM-judge claim enters the confirmatory scope.

Axiomatic properties and metric alignment. SHAP satisfies four axioms—dummy, linearity, symmetry, and efficiency—that guarantee attributions distribute the full prediction change $f(x) - \mathbb{E}[f]$ as uniquely fair marginal contributions. Perturbation-based fidelity metrics measure whether high-magnitude attributions correspond to large prediction drops under feature masking. These two perspectives are structurally compatible: both quantify how faithfully attribution magnitudes encode feature-level causal responsibility for the model output. SHAP's axiomatic guarantee implies that, conditional on its background distribution assumptions, its attribution rankings are optimally aligned with the feature-removal signal that fidelity metrics probe. This does not mean SHAP axioms certify correctness in all deployment contexts—the axioms hold for the model, not for any latent data-generative mechanism—but it does explain why SHAP consistently outperforms LIME on perturbation-based fidelity without requiring that LIME violates a formal criterion.

7.4 Internal and External Validity

Runtime comparisons are protocol-specific. SHAP combines TreeExplainer for tree models and KernelExplainer for non-tree models, whereas LIME is run with fixed local-surrogate settings (`num_features=10`, effective `num_samples=1000`, `kernel_width=3.0`). Accordingly, the reported cost gap reflects both method family and concrete implementation pathway. External generalization is partially supported by EXP3 (Section 5.9), which replicates SHAP's fidelity advantage over Anchors across Breast Cancer and German Credit datasets; SHAP-LIME comparisons and stability rankings remain bounded to the Adult cohort and may shift under different modalities, feature transformations, masking schemes, or hyper-parameter settings.

LIME stability is not a convergence artifact. A direct test of whether the near-zero LIME stability (cosine similarity mean ≈ 0.014, CV $= 86.2\%$) reflects insufficient perturbation samples was conducted by varying `num_samples` $\in \{500, 1000, 2000\}$ under otherwise identical conditions (RF model, seed 42, $N = 100$, `kernel_width=3.0`). Stability remained near-zero at all levels (0.011, 0.014, 0.016, respectively), while fidelity was essentially flat (0.452, 0.461, 0.459). Doubling or halving the sample budget neither stabilizes the explanations nor changes the fidelity ordering. The instability is therefore structural—it reflects inherent variability in LIME's local surrogate fitting process under Gaussian input perturbation, not an insufficient sample count that more compute would resolve. A companion `kernel_width` probe (0.25, 0.75, 3.0, 10.0) further shows that widths below 3.0 produce degenerate zero-weight explanations in this 103-dimensional preprocessed space, confirming that the reference setting is the minimum viable configuration rather than a conservative choice (see companion repository, `outputs/analysis/`).

Feature correlation and out-of-distribution masking. The Adult Income dataset contains moderate inter-feature correlations that affect the validity of perturbation-based fidelity metrics. Notably, `education` and `education-num` encode the same ordinal construct ($r = 1.0$ by construction); `occupation` and `education` show moderate Cramér's $V \approx 0.34$; and `hours-per-week` correlates with `occupation` and `workclass`. When these features are independently zeroed or replaced by reference values during masking—as in our fidelity and faithfulness-gap computations—the resulting masked instances may fall outside the training distribution. Under such out-of-distribution (OOD) inputs, tree-based model outputs can be unreliable extrapolations, and the correlation structure between masked and retained features is ignored. This can cause fidelity metrics to either overestimate (if the model is insensitive to OOD inputs) or underestimate (if the model reacts to implausible feature combinations) the true attribution quality. We cannot rule out that some portion of the SHAP–LIME fidelity gap reflects differential sensitivity to OOD masking artifacts rather than solely attribution accuracy. Studies using marginal versus conditional SHAP variants, or masking schemes that preserve feature dependencies, are needed to isolate this contribution.

7.5 Reproducibility and Operational Validity

All reported claims are traceable to seeded YAML configurations, per-run `results.json` artifacts, explicit exclusion of malformed or empty runs, and inferential exports in the companion repository. Figures are regenerated directly from the paired-cell Wilcoxon exports; no synthetic imputation is used at any stage. Before journal submission, the exact review snapshot should be frozen as a versioned release and archived under a version-specific DOI.

7.6 Future Work

Several limitations define natural extensions:

- **Dataset scope:** The primary paired benchmark focuses on the Adult tabular dataset; EXP3 partially extends fidelity replication to Breast Cancer and German Credit. Image, text, and graph domains remain untested and may exhibit different trade-offs.

- **Configuration space:** Both methods were run with default hyperparameter settings. A dedicated kernel-width probe (four values: 0.25, 0.75, 3.0, 10.0) reveals that values below 3.0 are degenerate in the 103-dimensional preprocessed Adult feature space—all weights collapse to zero—while `kernel_width` = 10.0 yields full instance coverage and improved stability (0.664 vs. near-zero) at a modest fidelity cost (0.441 vs. 0.518). These results confirm that the reference setting (3.0) is the minimum viable value in this space, not a conservative choice, and that the LIME instability observed in EXP2 persists across kernel-width settings rather than reflecting a calibration artifact. SHAP background-set sensitivity and cross-model kernel-width effects remain open.
- **Runtime heterogeneity:** The XGBoost exception shows that "LIME is faster" is not a universal rule and warrants tree-specific analysis.
- **Human-centered utility:** This paper does not establish human-grounded usefulness; validated user studies with task-outcome measures remain future work.
- **LLM-judge calibration:** EXP4 shows that ICC remains below 0.75 for all seven semantic dimensions; developing rubrics that achieve calibration-grade inter-rater agreement is an open challenge.
- **Taxonomy completeness:** The survey corpus is a working corpus rather than an exhaustive bibliometric sample; expanding it with systematic search protocols would strengthen the gap analysis.
- **Causal and counterfactual endpoints:** Integration of causal attribution methods and diverse counterfactual generation (Mothilal et al., 2020) remains an open follow-up.

8 Code and Artifact Availability

The manuscript, experimental code, and supporting materials are available in the public repository at `https://github.com/jonnabio/xai-eval-framework`, released under the MIT License. Artifacts needed to reproduce the empirical study include:

- EXP2 results: `experiments/exp2_scaled/results/`
- EXP2 inferential exports: `outputs/analysis/exp2_stats/` (includes `analysis_summary.json`, `wilcoxon_shap_lime_all_models.csv`, `paired_cells_shap_lime_all_models.csv`)
- EXP3 results: `experiments/exp3/results/`
- EXP4 ICC exports: `outputs/analysis/exp4_llm_evaluation/`
- analysis scripts: `scripts/run_exp2_statistical_analysis.py`, `scripts/generate_paper_b_figures.py`

The review corpus CSV underlying the taxonomy and gap analysis is also available in the companion repository. Before journal submission, the exact snapshot will be frozen as a versioned release with a DOI.

9 Conclusion

XAI evaluation is not one problem but a family of related measurement problems. The literature has developed useful metric families for fidelity, robustness, sparsity, and benchmark reproducibility, yet these families do not fully capture semantic adequacy or human-centered meaning. By organizing model-agnostic post-hoc evaluation along the axes of evaluation target, evidence source, quality property, and task context, this paper clarifies why metric disagreement is common and why layered evaluation is necessary.

The main practical implication is twofold. First, benchmark metrics and semantic evaluation should be treated as complementary rather than competing forms of evidence. The taxonomy makes explicit where each metric family gains strength and where it leaves blind spots. Second, the choice between LIME and SHAP is not a matter of one being universally better but of fit for purpose. Under the paired benchmark protocol, SHAP is the preferred choice when stability and fidelity-oriented quality dominate the objective, while LIME remains the pragmatic option when latency and compactness are the primary constraints.

Our paired analysis across 75 matched robustness cells shows consistent SHAP advantages in stability ($d_z = 3.00$), fidelity ($d_z = 4.82$), and faithfulness gap ($d_z = 2.63$), alongside systematic LIME advantages in sparsity and lower runtime for most model contexts. Cross-dataset replication (EXP3) confirms SHAP's fidelity advantage over Anchors across all 12 paired combinations on two independent datasets, strengthening the generalizability of the fidelity ranking. These results, read through the lens of the taxonomy, support a layered deployment architecture that separates user-facing low-latency explanation paths from high-assurance auditing workflows.

Acknowledgments and Disclosure of Funding

This work received no external funding. The study was self-funded by the author, and the author reports no competing interests. This draft was prepared from repository artifacts dated May 2026.

References

Alvarez-Melis, D. and Jaakkola, T. S. (2018). On the robustness of interpretability methods. In *ICML Workshop on Human Interpretability in Machine Learning (WHI 2018)*. https://arxiv.org/abs/1806.08049.

Adadi, A. and Berrada, M. (2018). Peeking inside the black-box: A survey on explainable artificial intelligence (XAI). *IEEE Access*, 6:52138–52160.

Adebayo, J., Gilmer, J., Muelly, M., Goodfellow, I., Hardt, M., and Kim, B. (2018). Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems*, volume 31, 9525–9536.

Agarwal, C., Ley, D., Krishna, S., Saxena, E., Pawelczyk, M., Johnson, N., Puri, I., Zitnik, M., and Lakkaraju, H. (2022). OpenXAI: Towards a transparent evaluation of model explanations. In *Thirty-Sixth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*. https://openreview.net/forum?id=MU2495w47rz.

Agarwal, C., Queen, O., Lakkaraju, H., and Zitnik, M. (2023). Evaluating explainability for graph neural networks. *Scientific Data*, 10:144.

Agrawal, K., El Shawi, R., and Ahmed, N. (2025). XAI-Eval: A framework for comparative evaluation of explanation methods in healthcare. *DIGITAL HEALTH*, 11:20552076251368045. https://doi.org/10.1177/20552076251368045.

Ali, S., Abuhmed, T., El-Sappagh, S., Muhammad, K., Alonso-Moral, J. M., Confalonieri, R., Guidotti, R., Del Ser, J., Díaz-Rodríguez, N., and Herrera, F. (2023). Explainable Artificial Intelligence (XAI): What we know and what is left to attain Trustworthy Artificial Intelligence. *Information Fusion*, 99:101805. https://doi.org/10.1016/j.inffus.2023.101805.

Arrieta, A. B., Díaz-Rodríguez, N., Del Ser, J., et al. (2020). Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 58:82–115.

Bansal, G., Wu, T., Vaughan, J., and Wallach, H. (2021). Does the whole exceed its parts? The effect of AI explanations on complementary team performance. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 1–16.

Burkart, N. and Huber, M. F. (2021). A survey on the explainability of supervised machine learning. *Journal of Artificial Intelligence Research*, 70:245–317. https://doi.org/10.1613/jair.1.12228.

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.

Buçinca, Z., Lin, P., Gajos, K. Z., and Glassman, E. L. (2020). Proxy tasks and subjective measures can be misleading in evaluating explainable AI systems. In *Proceedings of the 25th International Conference on Intelligent User Interfaces*, 454–464.

Canha, D., Kubler, S., Främling, K., and Fagherazzi, G. (2025). A functionally-grounded benchmark framework for XAI methods: Insights and foundations from a systematic literature review. *ACM Computing Surveys*, 57(12):1–40. `https://doi.org/10.1145/3737445`.

Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.

Covert, I., Lundberg, S., and Lee, S.-I. (2020). Understanding global feature contributions with additive importance measures. In *Advances in Neural Information Processing Systems 33*, 17212–17223. `https://arxiv.org/abs/2004.00668`.

Demsăr, J. (2006). Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30.

Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Giannotti, F., and Pedreschi, D. (2018). A survey of methods for explaining black box models. *ACM Computing Surveys*, 51(5):93:1–93:42. `https://doi.org/10.1145/3236009`.

Doshi-Velez, F. and Kim, B. (2017). Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv:1702.08608*. `https://arxiv.org/abs/1702.08608`.

Fok, R. and Weld, D. S. (2023). In search of verifiability: Explanations rarely enable complementary performance in AI-advised decision making. *AI Magazine*, 45(3):317–332.

Hase, P. and Bansal, M. (2020). Evaluating explainable AI: Which algorithmic explanations help users predict model behavior? In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 5540–5552. `https://doi.org/10.18653/v1/2020.acl-main.491`.

Gu, J., Jiang, X., Shi, Z., Tan, H., Zhai, X., Xu, C., Li, W., Shen, Y., Ma, S., Liu, H., Wang, S., Zhang, K., Wang, Y., Gao, W., Ni, L., and Guo, J. (2024). A survey on LLM-as-a-Judge. *arXiv preprint arXiv:2411.15594*. `https://doi.org/10.48550/arXiv.2411.15594`.

Haufe, S., Wilming, R., Clark, B., Zhumagambetov, R., Boubekki, A., Martin, J., and Panknin, D. (2026). Explainable AI needs formalization. *npj Artificial Intelligence*, 2:42. `https://doi.org/10.1038/s44387-026-00095-1`.

Hedstrom, A., Weber, L., Bareeva, D., Krakowczyk, D., Motzkus, F., Samek, W., Lapuschkin, S., and Hohne, M. M.-C. (2023). Quantus: An explainable AI toolkit for responsible evaluation of neural network explanations and beyond. *Journal of Machine Learning Research*, 24:1–11.

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.

Hooker, S., Erhan, D., Kindermans, P.-J., and Kim, B. (2019). A benchmark for interpretability methods in deep neural networks. In *Advances in Neural Information Processing Systems*, volume 32, 9737–9748.

Jacovi, A. and Goldberg, Y. (2020). Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness? In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 4198–4205.

Kaur, H., Nori, H., Jenkins, S., Caruana, R., Wallach, H., and Vaughan, J. W. (2020). Interpreting interpretability: Understanding data scientists' use of interpretability tools for machine learning. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–14. `https://doi.org/10.1145/3313831.3376219`.

Kadir, M. A., Mosavi, A., and Sonntag, D. (2023). Evaluation metrics for XAI: A review, taxonomy, and practical applications. In *Proceedings of the IEEE 27th International Conference on Intelligent Engineering Systems*, 111–124.

Kumar, I. E., Venkatasubramanian, S., Scheidegger, C., and Friedler, S. A. (2020). Problems with Shapley-value-based explanations as feature importance measures. In *Proceedings of the 37th International Conference on Machine Learning*, 5491–5500. `https://arxiv.org/abs/2002.11097`.

Kohavi, R. (1996). Scaling up the accuracy of Naive-Bayes classifiers: A decision-tree hybrid. In *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining*, 202–207.

Lipton, Z. C. (2018). The mythos of model interpretability. *Queue*, 16(3):31–57. `https://doi.org/10.1145/3236386.3241340`.

Lakens, D. (2013). Calculating and reporting effect sizes to facilitate cumulative science: A practical primer for t-tests and ANOVAs. *Frontiers in Psychology*, 4:863.

Longo, L., Brcic, M., Cabitza, F., Choi, J., Confalonieri, R., Del Ser, J., Guidotti, R., Hayashi, Y., Herrera, F., Holzinger, A., Jiang, R., Khosravi, H., Lecue, F., Malgieri, G., Páez, A., Samek, W., Schneider, J., Speith, T., and Stumpf, S. (2024). Explainable artificial intelligence (XAI) 2.0: A manifesto of open challenges and interdisciplinary research directions. *Information Fusion*, 106:102301. `https://doi.org/10.1016/j.inffus.2024.102301`.

Lundberg, S. M. and Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, 4765–4774.

Mohseni, S., Zarei, N., and Ragan, E. D. (2021). A multidisciplinary survey and framework for design and evaluation of explainable AI systems. *ACM Transactions on Interactive Intelligent Systems*, 11(3–4):1–45. `https://doi.org/10.1145/3387166`.

Mothilal, R. K., Sharma, A., and Tan, C. (2020). Explaining machine learning classifiers through diverse counterfactual explanations. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, 607–617.

Nauta, M., Trienes, J., Pathak, S., Nguyen, E., Peters, M., Schmitt, Y., Schlötterer, J., van Keulen, M., and Seifert, C. (2023). From anecdotal evidence to quantitative evaluation methods: A systematic review on evaluating explainability. *ACM Computing Surveys*, 55(13s):1–42. https://doi.org/10.1145/3583558.

Pawlicki, M., Pawlicka, A., Uccello, F., Szelest, S., D'Antonio, S., Kozik, R., and Choraś, M. (2024). Evaluating the necessity of the multiple metrics for assessing explainable AI: A critical examination. *Neurocomputing*, 602:128282. https://doi.org/10.1016/j.neucom.2024.128282.

Proszewska, M., Danel, T., and Rymarczyk, D. (2025). B-XAIC dataset: Benchmarking explainable AI for graph neural networks using chemical data. *arXiv preprint arXiv:2505.22252*.

Samek, W., Binder, A., Montavon, G., Lapuschkin, S., and Müller, K.-R. (2017). Evaluating the visualization of what a deep neural network has learned. *IEEE Transactions on Neural Networks and Learning Systems*, 28(11):2660–2673. https://doi.org/10.1109/TNNLS.2016.2599820.

Retzlaff, C. O., Angerschmid, A., Saranti, A., Schneeberger, D., Röttger, R., Müller, H., and Holzinger, A. (2024). Post-hoc vs ante-hoc explanations: xAI design guidelines for data scientists. *Cognitive Systems Research*, 86:101243. https://doi.org/10.1016/j.cogsys.2024.101243.

Slack, D., Hilgard, S., Jia, E., Singh, S., and Lakkaraju, H. (2020). Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 180–186. https://doi.org/10.1145/3375627.3375830.

Ribeiro, M. T., Singh, S., and Guestrin, C. (2016). Why should I trust you?: Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1135–1144.

Rong, Y., Leemann, T., Borisov, V., Kasneci, G., and Kasneci, E. (2022). A consistent and efficient evaluation strategy for attribution methods. In *Proceedings of the 39th International Conference on Machine Learning*, 18770–18795.

Wachter, S., Mittelstadt, B., and Russell, C. (2018). Counterfactual explanations without opening the black box: Automated decisions and the GDPR. *Harvard Journal of Law & Technology*, 31(2):841–887. https://doi.org/10.2139/ssrn.3063289.

Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5):206–215.

Sithakoul, S., Meftah, S., and Feutry, C. (2024). BEExAI: Benchmark to evaluate explainable AI. In *Explainable Artificial Intelligence*, 445–468. Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-63787-2_23.

Sokol, K. and Flach, P. (2020). Explainability fact sheets: A framework for systematic assessment of explainable approaches. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency*, 56–67.

Dua, D. and Graff, C. (2019). UCI Machine Learning Repository. University of California, Irvine, School of Information and Computer Science. http://archive.ics.uci.edu/ml.

Koo, T. K. and Li, M. Y. (2016). A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *Journal of Chiropractic Medicine*, 15(2):155–163. https://doi.org/10.1016/j.jcm.2016.02.012.

Ribeiro, M. T., Singh, S., and Guestrin, C. (2018). Anchors: High-precision model-agnostic explanations. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence*, 1527–1535. https://doi.org/10.1609/aaai.v32i1.11491.

Yeh, C.-K., Hsieh, C.-Y., Suggala, A. S., Inouye, D. I., and Ravikumar, P. (2019). On the (in)fidelity and sensitivity of explanations. In *Advances in Neural Information Processing Systems 32*, 10967–10978. https://arxiv.org/abs/1901.09392.

Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83.

Wilming, R., Budding, C., Müller, K.-R., and Haufe, S. (2022). Scrutinizing XAI using linear ground-truth data with suppressor variables. *Machine Learning*, 1–21.

Wilming, R., Kieslich, L., Clark, B., and Haufe, S. (2023). Theoretical behavior of XAI methods in the presence of suppressor variables. In *Proceedings of the 40th International Conference on Machine Learning*, 202, 37091–37107.

Wu, X., Kaushik, R., Li, W., Bauer, L., and Onoue, K. (2025). User perceptions vs. proxy LLM judges: Privacy and helpfulness in LLM responses to privacy-sensitive scenarios. *arXiv preprint arXiv:2510.20721*. https://doi.org/10.48550/arXiv.2510.20721.

Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E. P., Zhang, H., Gonzalez, J. E., and Stoica, I. (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems 36: Datasets and Benchmarks Track*. https://doi.org/10.48550/arXiv.2306.05685.

Zheng, X., Shirani, F., Chen, Z., Lin, C., Cheng, W., Guo, W., and Luo, D. (2025). F-Fidelity: A robust framework for faithfulness evaluation of explainable AI. In *The Thirteenth International Conference on Learning Representations (ICLR)*. https://openreview.net/forum?id=X0r4BN50Dv.

Zhou, S., Xie, W., Li, J., Zhan, Z., Song, M., Yang, H., Espinoza, C., Welton, L., Mai, X., Jin, Y., Xu, Z., Chung, Y.-H., Xing, Y., Tsai, M.-H., Schaffer, E., Shi, Y., Liu, N., Liu, Z., and Zhang, R. (2025). Automating expert-level medical reasoning evaluation of large language models. *npj Digital Medicine*, 9(1):34. https://doi.org/10.1038/s41746-025-02208-7.